ZIHO KANG

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RESEARCH INTERESTS

My research interests fall at the intersection of human factors, human-systems engineering, and data analytics. Especially, I develop algorithms to map human visual search patterns with human decision making processes that can be used to evaluate human performance, assess situation awareness, model decision making processes, inform the design of interfaces, and develop novel training methods. My recent research initiative involves multimodal analysis of eye tracking, haptic interactions, and brain activities to discover smart learning methods in multi-person virtual reality environment. I have recently received the NSF CAREER award to carry out my research initiatives.

Key words: human factors, human systems engineering, data analytics, human-computer interaction, eye tracking analytics, usability research, decision making, training interventions, simulation.
Applications: virtual reality, aerospace/aviation, weather, Deepwater Horizon explorations.

EDUCATION

- Ph.D., Industrial Engineering, Purdue University, West Lafayette, IN. May 2012.
  Dissertation: Dynamic sequence analysis of a decision making task of multielement target tracking and its usage as a learning method. Advisor: Dr. Steven Landry
- M.S., Industrial Engineering, Purdue University, West Lafayette, IN. Dec. 2006.
  Thesis: An improved optimization model for assigning empty freight cars in a railroad network
  Advisor: Dr. Jean-Philippe Richard.
- B.S., Industrial Engineering, Korea University, Seoul, South Korea. Mar. 2001.

PROFESSIONAL APPOINTMENTS

2014-present	Assistant Professor, School of Industrial and Systems Engineering, and Affiliate Professor (by courtesy), Department of Psychology, and Faculty member of Data Science and Analytics Program, University of Oklahoma, Norman, OK.
2013-2014	Post-doctoral Researcher, College of Computing and Informatics, Drexel University, Philadelphia, PA. Advisor: Dr. Ellen J. Bass.
2012-2013	Post-doctoral Research Associate, Department of Systems and Information Engineering, University of Virginia, Charlottesville, VA. Advisor: Dr. Ellen J. Bass
2004-2011	Research Assistant and Teaching Assistant, School of Industrial Engineering, Purdue University, West Lafayette, IN.
2003-2004	Engineer, Department of Research and Development of Trading Systems, Korea Stocks and Futures Exchange, Pusan, South Korea.
2001-2002	Engineer, Department of Research and Development of Enterprise Resource Planning Systems, Samsung Data Systems, Seoul, South Korea.
1997-1999	Honorable discharge as KATUSA Sergeant, Education Center,

Camp Humphreys, USASA, AREA III, South Korea.

SELECTED AWARDS AND HONORS

May 2019 Received **Excellent Presentation Award** in *Proceedings of the 5th International Conference on Education and Training technologies* (2019 ICETT), Seoul, South Korea.

May 2018 Received **Best University Research and Development Showcase Award*** in *Proceedings of the 50th Annual Offshore Technology Conference (OTC)*, Dallas, TX.
*Received as a collaboration team composed of Industrial and Systems Engineering, Petroleum and Geological Engineering, and Psychology at the University of Oklahoma.

Dec 2017 Nominated for the Institute of Industrial and Systems Engineers (IISE) Dr. Hamed K. Eldin Outstanding Early Career IE in Academia Award.

Sep 2017 Received **Excellent Presentation Award** in *Proceedings of the 2nd International Conference on Bio-engineering for Smart Technologies* (2017 BioSmart), Paris, France.

Oct 2016 Received **Andrew P. Sage Best Transaction Award** from the *IEEE Systems, Man, and Cybernetics Society*. Awarded for the best paper published in the *Transactions in Human-Machine Systems*.

Jan 2015 Received **Junior Faculty Fellowship Award** from the Office of the Vice President for Research at the University of Oklahoma.

Dec 2001 Received **Best New Recruit Award*** from Samsung Data Systems.
*Competed among approximately 600 recruits.

May 1999 Received **U.S. Army Recommendation Medal**.

Nov 1998 Received **Golden Tiger Decoration** upon passing the Primary Leadership Development Program from the U.S. Army.

PUBLICATIONS (Total: 64)

Underlined names indicate the students who published papers as my advisees (total of 15).

Peer-reviewed Journal Publications (Total: 17)

1. Argyle, E., Gourley, J. J., Kang, Z., Shehab, R. L. (2020, accepted). Investigating the Relationship between Eye Movements and Situation Awareness in Weather Forecasting. *Journal of Applied Ergonomics*.
2. Jeong H., **Kang. Z.**, and Liu Y. (2019). Driver glance behaviors and scanning patterns: Applying static and dynamic glance measures to the analysis of curve driving with secondary tasks. *Human Factors and Ergonomics in Manufacturing & Service Industries*, 29(6), 437-446.
3. Mandal, S. and **Kang. Z.** (2018). Using eye movement data visualization to enhance training of air traffic controllers: A dynamic network approach. *Journal of Eye Movement Research*, *11*(4), 1-20.
4. Salehi, S., Kiran, R., Jeon, J., and **Kang, Z.** (2018). Developing a cross-disciplinary, scenario-based training approach integrated with eye tracking data collection to enhance situational awareness in offshore oil and gas operations. *Journal of Loss Prevention in the Process Industries*, *56*, 78-94.
5. Khamaj, A., and **Kang, Z.** (2018). Usability evaluation of mobile weather hazard alert applications. *Industrial and Systems Engineering Review*, *6*(1), 21-40.
6. **Kang, Z.**, Dragoo, M.R., Yeagle, L.N., Shehab, R.L., Yuan, H., Ding, L., and West, S.G. (2018). Adaptive learning pedagogy of Universal Design for Learning (UDL) for multimodal training. *Journal of Aviation/Aerospace Education and Research*, *27*(1), 23-48.
7. Wilson, K.A., Heinselman, P.L., and **Kang, Z.** (2018). Comparing forecaster eye movements during the warning decision process. *Weather and Forecasting*, *33*(2), 501-521.

8. Argyle, E.M., Gourley, J.J., Ling, C., Shehab, R.L., and **Kang, Z.** (2017). Effects of display design on signal detection in flash flood forecasting. *International Journal of Human-Computer Studies*, *99*, 48-56.
9. **Kang, Z.** and Kim, K. (2017). Multimodal perception study on virtual 3D curved textures with vision and touch for interactive multimedia systems. *Multimedia Tools and Applications*, *77(2)*, 1-15.
10. **Kang, Z.**, Mandal, S., Crutchfield, J., Millan, A., and McClung, S.N. (2016) Designs and algorithms to map eye tracking data with dynamic multielement moving objects. *Computational Intelligence and Neuroscience*, *2016*, 1-18. (Appeared in the special issue of Advances in Eye Tracking Technology: Theory, Algorithms, and Applications)
11. **Kang, Z.** and Morin, T. (2016). Multi-attribute decision making under imperfect information and uncertain outcome of an unfair bidding game. *International Journal of Information Technology & Decision Making*, *15*(1), 63-81.
12. McClung, S.N. and **Kang, Z.** (2016). Characterization of visual scanning patterns in air traffic control. *Computational Intelligence and Neuroscience*, *2016*, 19-35. (Appeared in the special issue of Advances in Eye Tracking Technology: Theory, Algorithms, and Applications)
13. Wilson, K.A., Heinselman, P.L., Kuster, C.M., Kingfield, D.M., and **Kang, Z.** (2016). Forecaster performance and workload: Does radar update time matter? *Weather and Forecasting*, *32*, 253-274.
14. Wilson, K.A., Heinselman, P.L., and **Kang, Z.** (2016). Exploring applications of eye tracking in operational meteorology research. *Bulletin of the American Meteorological Society*, *97*(11), 2019-2025.
15. **Kang, Z.*** and Landry, S.J.***** (2015). An eye movement analysis algorithm for a multi-element target tracking task: Maximum transition-based agglomerative hierarchical clustering. *IEEE Transactions on Human-Machine Systems*, *45(1)*, 13-24.
    ***Received the Andrew P. Sage Best Transactions Paper Award.**
16. **Kang, Z.** and Landry, S.J. (2014). Using scanpaths as a learning method for a conflict detection task of multiple target tracking. *Human Factors: The Journal of Human Factors and Ergonomics Society*. *56*(1), 1150-1162.
17. **Kang, Z.** and Landry, S.J. (2014). Top-down approach for a linguistic fuzzy logic model. *Cybernetics and Systems: An International Journal*, *45*(1), 39-55.

Peer-reviewed Journal Papers under Writing (Total: 9)

1. Naeeri, S. and Kang, Z., Comparative analysis of fatigue in Novice and Expert Pilots' for Prolonged Multi-Flight Missions: Multimodal Analysis Approach Using Vigilance Test and Eye Tracking.
2. Jónsdóttir, A. A., Kang, Z., Sun, T., Mandal, S., Kim, J. Multimodal analysis in digital learning.
3. Plata, M. R. and Kang, Z., Universal Design for Learning for multimodal training.
4. Kang, Z., Mandal, S., Jeon, J., Graham, A., Salehi, S., Cokely, E. T. Prediction of Situation Awareness using Visual Scanpaths through Machine Learning: Application in Deepwater Horizon Operation, Human Factors: The Journal of Human Factors and Ergonomics Society.
5. Fraga, R. P., Kang, Z, and Mandal, S. Mixed methods approach of air traffic control strategies.
6. Khamaj, A. and Kang, Z. User-centered interface design and evaluation of time-critical services: Application in a mobile weather alert system.
7. Kang, Z. and Kim, K. Multimodal analysis of online healthcare systems.
8. Kang, Z. and Kim, K. User comprehension during interaction with online healthcare systems.
9. Kovelda, K. G. and Kang, Z. Perceived usability of mobile weather applications.

Peer-reviewed Conference Papers (Total: 17)

1. Naeeri, S., Mandal, S., and **Kang, Z.** (2019). Analyzing pilot fatigue for prolonged flight mission: Multimodal analysis approach using vigilance test and eye tracking. In *Proceedings of the Human Factors and Ergonomics Society 63rd Annual Meeting*, Oct. 28- Nov. 1, Seattle, WA.

2. Khamaj, A., **Kang, Z.**, and Argyle, E. (2019). Users' perceptions of smartphone weather applications' usability. In *Proceedings of the Human Factors and Ergonomics Society 63rd Annual Meeting*, Oct. 28- Nov. 1, Seattle, WA.
3. Yoon, D., **Kang, Z.**, Sun, F., and Kim, S. C. (2019). Audiences' attention to Youtube pre-roll advertising: Eye-tracking approach. In *Proceedings of the 2019 Annual Conference of American Academy of Advertising* (pp. 1-5), Mar. 28-31, Dallas, TX.
4. Kiran, R., Salehi, S., Jeon, J., and **Kang Z.** (2018a). Enhancing situation awareness and process safety in offshore drilling operations: Applications of eye-tracking system. In *Proceedings of the 50th Annual Offshore Technology Conference* (pp.1-18), Apr. 30-May 3, Houston, TX.
   Received OTC award for the Best University Research and Development Showcase.
5. Kiran, R., Salehi, S., Jeon, J., & **Kang, Z.** (2018b). Real-time eye-tracking system to evaluate and enhance situation awareness and process safety in drilling operations. In *Proceeding of the IADC/SPE Drilling Conference and Exhibition, Society of Petroleum Engineers* (pp.1-19), Mar. 6-8. Fort Worth, TX.
6. **Kang, Z.*** (2017). Real time eye movement analysis framework. In *Proceedings of the 2nd International Conference on Bio-engineering for Smart Technologies* (2017 BioSmart) (pp.1-4), Aug. 30-Sep. 1, Paris, France.
   ***Received Excellent Presentation Award.**
7. Naeeri, S., and **Kang, Z.** (2017). Analysis of Pilot's Visual Scanning Characteristics under Normal and Extreme Flight Conditions. In *Proceedings of the 6th Annual World Conference of the Society for Industrial and Systems Engineering* (pp.227-233), Oct. 12-13, Herndon, VA.
8. Khamaj, A., Shehab, R., and **Kang, Z.** (2017). Efficiency and distraction evaluations of motorcycles mounted infotainment systems and controls. In *Proceedings of the 6th Annual World Conference of the Society for Industrial and Systems Engineering* (pp.169-175), Oct. 12-13, Herndon, VA.
9. Ybarra, V.T., Dadmohammadi, Y., Salehi, S., **Kang, Z.**, Allan, J. N., Ramasubramanian, M., & Cokely, E. T. (2017). Drilling down on cognitive processes: Using process tracing with oil well operators (pp.1-5). In *Proceedings of the 2017 European Group of Process Tracing Studies in Judgment and Decision Making* (pp.1-5), Jun. 22-24, Galway, Ireland.
10. Yuan, H., Rippetoe, J., Ding, L., **Kang, Z.**, Shehab, R. L., and West. S. G. (2017). Universal Design for Learning in the framework of neuroscience-based education and neuroimaging-based assessment. In *Proceedings of the 2nd International Conference on Bio-engineering for Smart Technologies* (2017 BioSmart)(pp.1-4), Aug. 30-Sep. 1, Paris, France.
11. Wilson, K. A., Heinselman, P. L., and **Kang, Z.** (2016). Eye-tracking applications to assess impacts of phased array radar data on forecasters' cognitive processes. In *Proceedings of the 32nd Conference on Environmental Information Processing Technologies* (pp.1-6), American meteorological society, New Orleans, LA.
12. Mandal, S.*, **Kang, Z.**, and Millan, A. (2016). Data visualization of complex eye movements using directed weighted networks: a case study on a multi-element target tracking task. In *Proceedings of the Human Factors and Ergonomics Society 60th Annual Meeting* (pp.106-110), Washington D.C.
    ***Advisee, Mr. Mandal received Best Student Paper Award from the Aerospace Technical Group**.
13. Mandal, S. and **Kang, Z.** (2015). Eye tracking analysis using different types of Areas of Interest for multi-element moving objects: Results and implications of a pilot study in air traffic control. In *Proceedings of the Human Factors and Ergonomics Society 59th Annual Meeting* (pp.1515-1519), Los Angeles, CA.
14. **Kang, Z.** and Bass, E.J. (2014). Supporting the eye tracking analysis of multiple moving targets: Design concept and algorithm. *IEEE International Conference on Systems, Man, and Cybernetics* (pp.3191-3196), San Diego, CA.
15. **Kang, Z.**, Bass, E.J., and Lee, D.W. (2014). Air traffic controllers' visual scanning, aircraft selection, and comparison strategies in support of conflict detection. In *Proceedings of the 58th Annual Meeting of the Human Factors and Ergonomics Society* (pp.77-81), Chicago, IL.

16. **Kang, Z.** and Bass, E. J. (2013). Computationally tractable approaches to modeling the coverage of weather sensor networks, *IEEE International Conference on Systems, Man, and Cybernetics* (pp.4905-4912), Manchester, England.
17. **Kang, Z.** and Landry, S. J. (2010). Capturing and analyzing visual groupings of multiple moving targets in an aircraft conflict detection task using eye movements, In *Proceedings of the Human Factors and Ergonomics Society 54th Annual Meeting* (pp.1906-1910), San Francisco, CA.

Peer-reviewed Conference Papers under review (Total:4)

1. **Kang, Z.**, Fraga, R. P., Mandal, S., Crutchfield, J. M. (2020). Spatial-temporal cluster approach to discover visual scanning behaviors in a virtual reality airport tower. In *Proceedings of the Human Factors and Ergonomics Society 64th Annual Meeting*, Oct. 5- 9, Chicago, IL.
2. **Kang, Z.**, Jeon, J., Salehi, S., (2020). Eye tracking analytics for virtual reality training, In *Proceedings of the Human Factors and Ergonomics Society 64th Annual Meeting*, Oct. 5- 9, Chicago, IL.
3. Crutchfield, J. M., Willams, B., Post, D. P., Bass, E. J., Landry, S. J., **Kang, Z.**. (2020). Human factors methods and approaches to increase safety in near future air traffic management environment. In *Proceedings of the Human Factors and Ergonomics Society 64th Annual Meeting*, Oct. 5- 9, Chicago, IL.
4. Fraga, R. P., Reddy, Y. P., **Kang, Z.**, Izzetoglu, K. (2020). Multimodal analysis using neuroimaging and eye movements to assess cognitive workload. In *Proceedings of the 22nd International Conference on Human-Computer Interaction*, Jul. 19-24, Copenhagen, Denmark.

Conference Proceedings, Magazines, Posters, and Other Published Manuscripts Not Critically Reviewed (Total: 23)

1. Mandal, S. and **Kang, Z**. (2019). Visual scanpath approximation for training enhancement: A spatial-temporal clustering approach. In *2019 Symposium on Artificial Intelligence and Machine Learning at the University of Oklahoma*, Sep. 25-27, Norman, OK.
2. **Kang, Z.*** (2019). Human error prevention training using eye tracking technology. In *2019 5th International Conference on Education and Training Technologies*, May 27-29, Seoul, S. Korea. (Abstract)
   ***Received Excellent Presentation Award**
3. **Kang, Z.** (2019). Improving learning performance by implementing Universal Design for Learning (UDL) pedagogy in higher education. In *2019 5th International Conference on Education and Training Technologies*, May 27-29, Seoul, S. Korea. (Abstract)
4. **Kang, Z.**, Dyer, J. W., West, S. G., Fraga, R. P.*, Mandal, S., Egwu, U. K., and McClung. S. N. (2019). Characterization of air traffic controllers' visual search patterns and control strategies. *Technical Training and Human Performance* (TTHP) *Meeting*, Federal Aviation Administration Center of Excellence, Ohio State University, Columbus, OH, Apr. 17, 2019. (Poster competition)
   ***Advisee, Mr. Fraga, received Best Poster Presentation Award (two consecutive years)**
5. **Kang, Z.**, Shehab, R. L., Ding, L., Yaun, H., West, S. G., Fraga, R. P., Plata, M. R., Alhashim, A., Annadi, R. R., Dragoo, M. R., and Yeagle, L. N. (2019). Universal Design for Learning and Multimodal Training (poster exhibition). *Technical Training and Human Performance* (TTHP) *Meeting*, Federal Aviation Administration Center of Excellence, Ohio State University, Columbus, OH, Apr. 17, 2019. (Poster competition)
6. **Kang, Z.**, Dyer, J. W., West, S. G., Fraga, R. P., Mandal, S., Egwu, U. K, and McClung. S. N. (2018). Characterization of air traffic controllers' visual search patterns and control strategies (poster exhibition). In *Proceedings of the 2018 Interservice/ Industry Training, Simulation and Education Conference (I/ITSEC)*, Nov. 26-30, Orlando, FL. (Poster exhibition)

7. **Kang, Z.**, Shehab, R. L., Ding, L., Yaun, H., West, S. G., Dragoo, M. R., Yeagle, L. N., Fraga, R. P., and Rippetoe, J. (2018). Adaptive learning pedagogy of Universal Design for Learning (UDL) for multimodal training (poster exhibition). In *Proceedings of the 2018 Interservice/ Industry Training, Simulation and Education Conference (I/ITSEC)*, Nov. 26-30, Orlando, FL. (Poster exhibition)
8. **Kang, Z.**, Jeon, J., Salehi., S., Cokely, E. T., and Ybarra, V. T. (2018). Situation awareness assessment through protocol and eye tracking analyses. In *Proceedings of the 2018 Institute of Industrial and Systems Engineers (IISE) Annual Conference* (pp.1-5), May 19-22, Orlando, FL. (Conference paper)
9. Mandal, S.*****, and **Kang, Z.** (2018). Analyzing eye movement data for dynamic scenarios: Information theory and network science based integrated framework. *2nd Annual Gallogly College of Engineering Poster Fair*, University of Oklahoma, Nov. 10, Norman, OK. (Poster competition)
   ***Advisee, Mr. Mandal, received Graduate Poster Competition Award (1st place).**
10. Naeeri, S. and **Kang, Z.** (2018). Exploring the relationship between pilot's performance and fatigue when interacting with cockpit interfaces. In *Proceedings of the 2018 Institute of Industrial and Systems Engineers (IISE) Annual Conference* (pp.1-5), May 19-22, Orlando, FL. (Conference paper)
11. Fraga, R. P., **Kang, Z.**, and Mandal, S. (2018). Characterization of air traffic controllers' visual search patterns and control strategies. In *Proceedings of the 2018 International Society for Engineers and Researchers* (pp.1-5), Jun. 18-19, Seoul, S. Korea. (Conference paper)
12. Yeagle, N. L. and **Kang, Z.** (2018). Universal Design for Learning (UDL) for STEM and higher education: Characterizations and applicability. In *Proceedings of the 2018 International Society for Engineers and Researchers* (pp.1-5), Jun. 18-19, Seoul, S. Korea. (Conference paper)
13. **Kang, Z.**, Dyer, J., West, S. G., Mandal, S., Fraga, R. P.*****, McClung, S., and Egwu, U. K. (2017). Characterization of visual scanning patterns and aircraft control strategies for efficient and effective training. Poster session in *Solutions for Operational Aviation Research (SOAR) Q2 meeting*, Federal Aviation Administration Center of Excellence, Apr. 3-5, Philadelphia, PA. (Poster competition)
    ***Advisee, Mr. Fraga, received Best Poster Presentation Award along with scholarship**
14. **Kang, Z.**, Shehab, R. L., Ding, L., Yaun, H., West, S. G., Dragoo, M. R., Yeagle, L. N., Fraga, R. P., and Rippetoe, J. (2017). Universal Design for Learning and Multimodal Training. Poster session in *Solutions for Operational Aviation Research (SOAR) Q3 meeting*, Federal Aviation Administration Center of Excellence, Apr. 3-5. Philadelphia, PA. (Poster competitiion)
15. **Kang, Z.**, Dragoo, M.R., Yeagle, L.N., Shehab, R. L., Yuan, H., Ding, L., and West, S. G. (2017). Adaptive learning pedagogy in Universal Design for Learning and multimodal training. In *Proceedings of the 2017 National Training Aircraft Symposium (NTAS)*(pp.1-12), Aug. 14-16. Daytona Beach, FL. (Conference paper)
16. **Kang, Z.**, Mandal, S., and Dyer, J. (2017). Data visualization approaches in eye tracking to support the learning of air traffic control operations. In *Proceedings of the 2017 National Training Aircraft Symposium (NTAS)*(pp.1-10), Aug. 14-16. Daytona Beach, FL. (Conference paper)
17. **Kang, Z.*** and Landry S. J.***** (2017). An eye movement analysis algorithm for a multi-element target tracking task: Maximum transition-based agglomerative hierarchical clustering. *IEEE Systems, Man, and Cybernetics Magazine*, *2*, 52-53. (Magazine)
    ***Extended abstract published in magazine as a recipient for the Andrew P. Sage Best Transactions Paper Award**.
18. **Kang, Z.**, Shehab, R. L., Ding, L., Yaun, H., West, S. G., Dragoo, M. R., Yeagle, L. N., Fraga, R. P., and Rippetoe, J. (2017). Universal Design for Learning and Multimodal Training. *Solutions for Operational Aviation Research (SOAR) Q4 meeting*, Federal Aviation Administration Center of Excellence, FAA headquarters, Jun. 11-15, Washington D.C. (Poster exhibition)
19. **Kang, Z.**, Dyer, J., West, S. G., Mandal, S., Fraga, R. P., McClung, S., and Egwu, U. K. (2017) Characterization of visual scanning patterns and aircraft control strategies for efficient and effective training. *Solutions for Operational Aviation Research (SOAR) Q4 meeting*, Federal Aviation Administration Center of Excellence, FAA headquarters, Jun. 11-15, Washington D.C. (Poster exhibition)

20. Khamaj, A. and **Kang, Z.** (2017) Usability evaluation of mobile weather alert applications. *1st Annual Gallogly College of Engineering Poster Fair*, University of Oklahoma, Nov. 10, Norman, OK. (Poster competition)
21. Mandal, S. and **Kang, Z.** (2017, November) Visualizing eye movement data in dynamic object tracking tasks. *1st Annual Gallogly College of Engineering Poster Fair*, University of Oklahoma, Nov. 10, Norman, OK. (Poster competition)
22. Dadmohammdi, Y., Salehi, S., Kiran, R., Jeon, J., **Kang, Z.**, Cokely, E.T., and Ybarra, V.T. (2017) Integrating human factors into petroleum engineering's curriculum: Essential training for students, In *Proceedings of the Society of Petroleum Engineers (SPE) Annual Technical Conference and Exhibition* (pp.1-14), Oct. 9-11, San Antonio, TX. (Conference paper)
23. Ybarra, V.T., Dadmohammadi, Y., Salehi, S., **Kang, Z.**, Allan, J. N., Ramasubramanian, M., & Cokely, E. T. (2017), In *Proceedings of the 2017 Offshore Technology Conference*, May 1-4, Houston, TX. (Poster exhibition)

Technical Reports (Total: 7)

1. **Kang, Z.**, Shehab, R. L., Ding, L, Yuan, H., Fraga, R. P., Yeagle, L. N., Alhashim, A., Plata, M. R., Dragoo, M. R. (2019). Universal Design for Learning (UDL) and multimodal training. pp. 1-177. Federal Aviation Administration.
2. Salehi, S., **Kang, Z.**, Cokely, E., Muhammed, R. A., Kiran, R., Jeon, J., Ybarra, V. (2018). Virtual reality offshore operations training infrastructure: Enhancing expert containment, decision making, and risk communications. pp. 1-30. National Academy of Sciences.
3. **Kang, Z.**, Dyer, J. W., West, S. G., Fraga, R. P., Mandal, S., Egwu, K, and McClung. S. N. (2018). Characterization and application of air traffic controllers' visual search patterns and control strategies for efficient and effective training. pp. 1-282. Federal Aviation Administration.
4. **Kang, Z.** and Mandal, S. (2017). Development of design concepts and algorithms in eye tracking research for a multi-element objects tracking task to support human performance analysis. pp. 1-27. Federal Aviation Administration.
5. **Kang, Z.** (2017). Development of a mixed methods approach to inform the design of new learning methods for a complex task: An application in multiple objects tracking task. Center for Research Program Development and Enrichment at the University of Oklahoma.
6. Bass, E. J., **Kang, Z.**, Philips, B., Lyons, E., Bajaj, A., Junyent, F., and Beauchamp, R. (2014). Healthcare Systems Integrated Warning Assessment (HIWA) final report. pp. 1-95. North Central Texas Trauma Regional Advisory Council.
7. Richard, J. P., **Kang, Z.**, and Narisetty, A. K. (2006). SOM v2.0: A simplified optimization model for empty car assignments. pp. 1-80. Union Pacific Company.

PRESENTATIONS (EXCLUDING CONFERENCE PRESENTATIONS), PANEL DISCUSSIONS, WEBINARS, AND EXHIBITIONS (Total: 39)

Invited Presentations (Total 11):

1. "**How do we observe information of interest? - Attention research in marketing and training**," Price College of Business, University of Oklahoma, Jan. 24, 2020, Norman, OK.
2. "**Data Analytics in Human Factors Research**," School of Industrial and Systems Engineering, University of Washington, Oct. 29, 2019, Seattle, WA.
3. "**Data analytics in Human-Technology Frontier**," *International Research Seminar*, Department of Industrial Management Engineering, Korea University, Jun. 11, 2019, Seoul, South Korea.

4. "**Data analytics in Human-Technology Frontier**," *International Research Seminar*, Department of Information and Telecommunications Engineering, Incheon National University, Jun. 5, 2019, Seoul, South Korea.
5. "**Data analytics in Human-Technology Frontier**," *International Research Seminar*, Department of Industrial Engineering, Hongik University, May 30, 2019, Seoul, South Korea.
6. "**Human role in complex systems: Trust in automation, information reliance, and information overload**," *Human Systems Integration Summit*, Mike Monroney Aeronautical Center, Federal Aviation Administration, Apr. 11, 2019, OKC, OK.
7. "**Advancements of eye movement analysis methodologies for a dynamic task**," *Tech Talk.* Air Force Research Laboratory, Department of Defense, Oct. 18, 2018, Austin, TX.
8. "**Human systems integration: Eye tracking for training**," *Human Systems Integration Summit*, Mike Monroney Aeronautical Center, Federal Aviation Administration, May 23, 2018, OKC, OK.
9. "**Eye tracking data analysis algorithms and applications**," Department of Information and Industrial Engineering at Yonsei University, Jul. 21, 2015, Seoul, S. Korea.
10. "**Eye tracking data analysis methods for a tracking task of multiple moving targets**," *Federal Aviation Administration Collaborative Research Meeting*, Civil Aerospace Medical Institute, Mike Monroney Aeronautical Center, Federal Aviation Administration, Oct. 10, 2014, OKC, OK.
11. "**Analysis of driving behaviors using eye movements**," *1st Hyundai Global Top Talent Forum*, 2011 World Motor Auto Show, Jan. 10, 2012, Detroit.

Other Presentations (Total 18):

1. "**Role of human factors in human-technology frontier**," *Human Factors and Ergonomics Joint Chapter Meeting between University of Oklahoma and Ohio State University*, Norman, OK, Oct. 1, 2019.
2. "**Eye tracking research in human factors,**" *Human Factors and Ergonomics Society OSU Chapter Meeting*, Ohio State University, Columbus, OH, Apr. 17, 2019.
3. "**Characterization of visual scanning patterns and aircraft control strategies for training**," *Technical Training and Human Performance* (TTHP) *Meeting*, Federal Aviation Administration Center of Excellence, Ohio State University, Columbus, OH, Apr. 17, 2019.
4. "**Universal Design for Learning (UDL) and multimodal training**," *Technical Training and Human Performance* (TTHP) *Meeting*, Federal Aviation Administration Center of Excellence, Ohio State University, Columbus, OH, Apr. 17, 2019.
5. "**Situation awareness assessment through eye movement analysis in Deepwater Horizon operations**," *Research meeting with SeaDrill Limited*, Norman, OK. Oct. 31, 2018. (Presentation)
6. "**Eye tracking data analytics**," *Human Factors and Ergonomics OU Chapter Big Meeting*, Norman, OK, Nov. 1, 2017.
7. "**Data visualization methods in eye movement research**," *OU ISE graduate seminar*, Norman, OK, Oct. 6, 2017.
8. "**Q4 Characterization of visual scanning patterns and aircraft control strategies for training**," *Solutions for Operational Aviation Research (SOAR) Q4 meeting*, Federal Aviation Administration Center of Excellence, FAA headquarters, Washington D.C., Jun. 15, 2017.
9. "**Q3 Universal Design for Learning (UDL) and multimodal training**," *Solutions for Operational Aviation Research (SOAR) Q3 Meeting*, Federal Aviation Administration Center of Excellence, Embry-Riddle Aeronautical University, Daytona, FL., 1 Mar. 2017.

10. "**Q3 Characterization of visual scanning patterns and aircraft control strategies for training**," *Solutions for Operational Aviation Research (SOAR) Q3 Meeting*, Federal Aviation Administration Center of Excellence, Embry-Riddle Aeronautical University, Daytona, FL., Mar. 1, 2017.
11. "**Q2 Universal Design for Learning (UDL) and multimodal training**," *Solutions for Operational Aviation Research (SOAR) Q2 Meeting*, Federal Aviation Administration Center of Excellence, University of Oklahoma, Norman, OK, Dec. 1, 2017.
12. "**Q2 Characterization of visual scanning patterns and aircraft control strategies for training**," *Solutions for Operational Aviation Research (SOAR) Q4 Meeting*, Federal Aviation Administration Center of Excellence, University of Oklahoma, Norman, OK, Dec. 1, 2017.
13. "**Human factors professions and introduction to eye tracking**," *Human Factors and Ergonomics Society OU Chapter Meeting*, Norman, OK. Sep 15, 2016.
14. "**Visual scanning analysis**" *Research meeting with Laureate Institute for Brain Research*, Norman, OK. Jan 6, 2016.
15. "**Eye tracking analysis: A demonstration**," *Human Factors and Ergonomics Society OU Chapter Meeting*, Norman, OK. Mar. 12, 2015.
16. "**Towards big data analytics: Understanding database management systems**," *Special seminar for IIE members and ISE students*, Norman, OK, Mar 9, 2015.
17. "**Big data analytics: Visual scan path analysis**," *Big data analytics meeting*, Center for Research Program Development and Enrichment, Norman, OK, Feb 20, 2015.
18. "**Eye tracking research: Visual groupings and hierarchical clustering**," *OU ISE graduate seminar*, Norman, OK, Oct. 22, 2014.

Invited Panel Discussions (Total 4):

1. "**Transitioning through academia: Graduate students, postdocs, and tenure track faculty**," In Proceedings of the Human Factors and Ergonomics Society 60th Annual Meeting, Oct. 28- Nov. 1, Seattle, WA.
2. "**Human factors research roadmap**," Center of Excellence for Technical Training and Human Performance Workshop, Federal Aviation Administration Center of Excellence, Jun. 6, 2018, Norman, OK.
3. "**Technical training roadmap**," Center of Excellence for Technical Training and Human Performance Workshop, Federal Aviation Administration Center of Excellence, Jun. 5, 2018, Norman, OK.
4. "**Improving the mission readiness of employees and safety of the National Airspace System (NAS)**," Solutions for Operational Aviation Research (SOAR) Q2 Meeting, Federal Aviation Administration Center of Excellence, Apr. 4, 2018, Philadelphia, PA.

Invited Webinars (Total 1):

1. "**Data visualizations in eye tracking research**," *Tobii Pro Research Spotlight Seminar Series*, Tobii Pro North America, Jun. 1, 2017.

Other Webinars (Total 2):

1. "**Future research topics on visual search and control strategies**," *Solutions for Operational Aviation Research (SOAR) Meeting*, Federal Aviation Administration. Jun. 7, 2018.
2. "**Characterization and application of air traffic controllers' visual search patterns and control strategies for efficient and effective training**," *Solutions for Operational Aviation Research (SOAR) Project Closeout Briefing*, Federal Aviation Administration. Apr. 24, 2018.

Invited Exhibitions (Total 3):

1. "**Universal Design for Learning (UDL) and multimodal training**," *2018 Interservice/Industry Training, Simulation and Education Conference (I/ITSEC)*, Nov. 26-30, 2018. Orlando, FL. (Invited by Federal Aviation Administration Center of Excellence)
2. "**Characterization of air traffic controllers' visual search patterns and control strategies**," *2018 Interservice/Industry Training, Simulation and Education Conference (I/ITSEC)*, Nov. 26-30, 2018. Orlando, FL. (Invited by Federal Aviation Administration Center of Excellence)
3. "**Capturing and analyzing visual groupings of multiple moving targets in an aircraft conflict detection task using eye movements**," *Air Traffic Control Association (ATCA) 55th Annual Conference and Exposition*, Oct. 24-27, 2010, Alexandria, V.A. (Invited by the President of ATCA)

TEACHING AND ADVISING

Instructor Activities (Total: 3 courses)

Cognitive Engineering and Decision Making (ISE 5813)***: The course integrates human factors and systems engineering. The core contents are analysis, design, and evaluation of human behaviors and decision making processes in simple and complex systems. Detailed teaching contents include, but are not limited to, methodologies to evaluate human performance and model human decision making processes including human error, signal detection theory, utility theory, information theory, information processing model, information display, perception, attention, workload, memory, and comprehension. (Teaching period: **2015-current**)

***Course name was modified to "Advanced Human Factors and Ergonomics" in 2019.*

Class evaluations (out of 5.0 and rounded to 1 decimal place):

Year (Sp.)	2015	2016	2017	2018	2019	Mean
Enrollments*	12	24	27	27	22	-
Individual*	3.4	4.6	4.3	4.8	4.5	4.6
Department	4.3	4.1	4.3	4.2	4.4	4.3
College	4.3	4.2	4.2	4.1	4.2	4.2

***Enrollments have **doubled** from 12 in 2015 up to 27 in 2018.*

Evaluation has **increase from 3.4 in 2015 up to 4.8 in 2018.*

Selected students' comments:

1. "Love Dr. Kang! He has good teaching methods and you can tell he's passionate about his work. Always makes time to meet with students even outside normal office hours."
2. "Dr. Kang has grown to be one of the best and influential professors I've had during my time at OU. Regarding his class, expectations are clear and fair. What is needed to be known is clear and the supplementary material really enforced these key concepts. He is excited and more than willing to help during office hours, more than any other professor I have had. As a student I feel much more inclined to work hard for a professor who I know works hard to teach me. I took this course 1) because Dr. Kang was teaching it and 2) I thought the subject matter was interesting. I did not initially expect to rely on and use these concepts in my professional career, but during lectures I was able to make parallels and connections to these applications in the field I am heading toward."

Data Driven Decision Making II (ISE 4853 and ISE 5853)***: Both ISE 4853 and ISE 5853 are taught together in a single classroom. This course introduces the students to design experiments and apply statistical analysis through industrial and systems engineering-related topics integrated with

human performance measures. Detailed elements of the experimental methodologies that are taught include, but are not limited to, developing applied research problems, determining meaningful variables and measurements, formulating hypotheses and associated mathematical models, collecting and analyzing data, using sophisticated software that supports statistical analyses, developing professional presentation slides and reports. Specific materials include, but are not limited to, mathematical modeling of complex experiment designs, parametric/non-parametric statistical methods, logistic regression, ANCOVA, database management system, scientific writing, and the use of modern tools such as SAS, R, and Microsoft Access. (Teaching period: **2015-current**)

Class evaluations (out of 5.0 and rounded to 1 decimal place):

Year (Fa.)	2015	2016	2017	2018	2019	Mean
Enrollments*	45	53	77	93	57	-
Individual	3.2	4.0	4.4	3.9	4.3	4.0
Department	4.1	4.3	4.0	4.1	4.3	4.1
College	4.1	4.2	4.2	4.1	4.2	4.1

***Enrollments have **doubled** from 45 in 2015 to 93 in 2018.*
Evaluation has **increase from 3.2 in 2015 up to 4.4 in 2018.*

Undergraduate student paper competition winners using the DDDM II course project reports:

1. Team composed of Lauren Yeagle, Mattlyn Dragoo, Mathew Pourfakhrai, Marissa Bailey, and Julius Ta won 1st place at the IISE South Central Regional Conference **(**held at Texas Tech University in 2017). Project title: Usability analysis of online weather interfaces.
2. Team "Woohoo" (Leslie Barnes, Jack Appleyard, Akash Amin, Rodney Gottschalk, and Chisom Ibekwe) won 3rd place at the IISE South Central Regional Conference **(**held at Texas A&M University in 2018). Project title: Differences in Risk Perception For Different Icon Array Formats.
3. In 2019, the students did not compete, therefore, I have been working with the undergraduate student teams so that they could submit either a conference or journal paper.

Selected students' comments:

1. "Very professional with a no nonsense kind of attitude which is needed for this program. Dr. Kang killed it with DDDM2 with his great lectures and supplementing material. Perfect guy to teach this class"
2. "Amazing professor. His lecture slides are very clear and even when you have to miss a class, reading the lecture helps you understand the materials really well. He cares about his students and want them to succeed."
3. "When I first started this class I wasn't quite sure if I was going to learn anything, but right now I am glad I had Dr. Kang as my teacher. I have never been good at right technical, or even at writing at all. He was able to truly guide me with helping me prep properly with his slides and lectures. I also believe he was the teacher that put the most amount of effort into making his slides and designing his lectures. He is honestly one of the top 2 professors in ISE."
4. "The class slides were some of the best I've seen. The course felt like 3 different classes, and I initially didn't like that the homework assignments weren't problems for what we did in class. In the end, I really liked that the assignments broke up the semester project to make it more manageable. It was significantly easier to do it with the different deadlines."

<u>Systems Analysis using Simulation (ISE 4663) and Simulation I (ISE 5663)</u>***: Both ISE 4663 and Simulation 5663 are taught together in a single classroom. This course introduces the students the art and science of collecting and analyzing data for system analysis, technologies to design, implement, run, and animate simulation models, methods to analyze and interpret results, and the opportunities to design and analyze simulation models of various systems. Specific materials include, but are not limited to, stochastic simulation theories and applications (including manufacturing systems, service systems, human-integrated systems), terminal vs. steady state systems, queuing theory, distributions, non-stationary Poisson process, scheduling, statistical analysis of performance measures, and use of the Arena simulation software. (Teaching period: **2017-current**)

Class evaluations (out of 5.0 and rounded to 1 decimal place):

Year (Fa.)	2017	2018	2019	Mean
Enrollments*	74	94	61	-
Individual	4.0	3.8	4.4	4.1
Department	4.0	4.1	4.3	4.1
College	4.2	4.1	4.2	4.2

Lab evaluations (out of 5.0 and rounded to 1 decimal place):

Year (Fa.)	2017	2018	2019	Mean
Enrollment*	74	94	61	-
Individual	4.2	4.0	4.5	4.2

Enrollments have *doubled* from approximately 45 in 2015 to 94 in 2018.

Selected students' comments:

1. "Kang's slides are AMAZING. The slides he makes for the course are very detailed and easily understood. Towards the end of class Dr. Kang would bring paper and encourage students to participate by drawing process examples."
2. "Dr. Kang gave a lot of effort during and out of class and it shows on the material itself. 10/10 would take this class again."
3. "Dr. Kang Helped me get through a Traumatic Brain Injury and helped me understand how to regain my confidence. I am greatly appreciative."
4. "Kang is the best professor I've had during my time at ou. He actually tries to teach the students no matter how complicated the subject. He will walk you through everything and he even created his own way to teach the subject that made it easy to understand rather than following the book word for word like 95% of professors. He should be paid very well for his time. God bless people like Dr. Kang."

Ph.D. Advising: Ph.D. Dissertation Committee Chair (Total: 4)

Note: OU stands for University of Oklahoma.

1. Salem Naeeri*, Industrial and Systems Engineering, OU, May 2020.
2. Saptarshi Mandal*, Industrial and Systems Engineering, OU, May 2020.
3. Abdulrahman Khamaj*****, Industrial and Systems Engineering, OU, Dec 2019.
   Thesis: Towards an enhanced user experience with smart phone weather alert applications: Usability and user-centered design approaches.
4. Elizabeth M. Argyle*****, Industrial and Systems Engineering, May 2016.
   Thesis: Supporting situation awareness and decision making in weather forecasting.
   Co-advised with Dr. Randa L. Shehab.

** Salem received the Robberson Conference Presentation & Creative Exhibition Travel Grant from OU. (awarded $1K).*
** Saptarshi won the Gallogly College of Engineering Dissertation Excellence Award. (awarded $5K).*
** Abdulrahman is currently a Tenure-track Assistant Professor at Jazan University in Saudi Arabia.*
** Elizabeth is currently a Research Fellow at the Institute for Aerospace Tech. at the U. of Nottingham.*

Ph.D. Advising: Ph.D. Dissertation Committee Member (Total: 4)

1. Aksay Gaur, Department of Computer Science, OU, expected Dec. 2021.
2. Vera Luiza Wendler Bosco, Industrial and Systems Engineering, OU, expected Dec. 2021.
3. Katie A. Wilson*****, Department of Meteorology, OU, Dec. 2018 .
   Thesis: Forecaster warning decision making with rapidly-updating radar data.
4. Chinedum Peter Ezeakacha, School of Petroleum and Geological Engineering, OU, Dec. 2017.
   Thesis: Dynamic drilling fluids loss and filtration: impact of dynamic wellbore conditions and wellbore strengthening implications.

** Katie is currently a research associate at the National Weather Service, Norman, OK.*

M.S. Advising: M.S. Thesis Committee Chair (Total: 8)

1. Malissa Rosa Plata, Industrial and Systems Engineering, OU, expected May 2020.
2. Ricardo P. Fraga*****, Industrial and Systems Engineering, OU, expected May 2020.
3. Elizabeth N. Blanche*****, Industrial and Systems Engineering, OU, Dec 2018.
   Thesis: Human-computer interface design in human factors: Weapons drop.
4. Jiwon Jeon*****, Data Science and Analytics Program, OU, May 2018.
   Thesis: Analysis of eye tracking data to measure situational awareness in offshore drilling operations.
5. Lucas Cezard, Industrial and Systems Engineering, OU, Dec 2017.
   Thesis: Prediction of user intentions when operating with the cursor.
6. Lucas Mozoni*****, Industrial and Systems Engineering, OU, Dec 2017.
   Thesis: Analysis of the relationships between eye tracking information and text comprehension levels in healthcare.
7. Sarah N. McClung*****, Electrical and Computer Engineering, OU, Aug. 2017.
   Thesis: Mode reconfigurable bandpass filters with liquid metal enablement.
   Co-advised with Dr. Hjalti H. Sigmarsson.
8. Saptarshi Mandal*****. Industrial and Systems Engineering, OU, Dec. 2016.
   Thesis: Analyzing eye tracking data in multielement moving target tracking: Directed weighted network analysis.

** Elizabeth is a flight systems engineer at Tinker Air Force Base in OKC, OK.*
** Jiwon received the **Best University Research and Development Showcase Award** in Proceedings of the 50th Annual Offshore Technology Conference. She is currently a data analyst working in Dallas, Texas.*
** Lucas is a consultant working in New York, NY. Lucas won the 2017 **Best M.S. Thesis** award.*
** Sarah is currently a consultant of her family business located in OKC, OK.*
** Saptarshi is currently studying with me as a Ph.D. student. Saptarshi won the **Best Student Paper Award** from the Aerospace TG in the Proceedings of the 2015 Human Factors and Ergonomics Society 60th Annual Meeting, Washington D.C., won **1st place in the Graduate Poster Competition** at the University of Oklahoma, and won the 2016 **Best M.S. Thesis** award.*
** Ricardo won the **Best Poster** award during the FAA SOAR Meeting, Philadelphia, PA in 2018 and the FAA TTHP Meeting, Columbus, OH in 2019. He is planning to continue his Ph.D. with me upon receiving his M.S. degree.*

M.S. Advising: Thesis Committee Member (Total: 9)

1. Lauren N. Yeagle, Data Science and Analytics Program, OU, Dec. 2018.
   Thesis: Measuring and optimizing influence for resilient community networks.

2. Gowtham Talluru, Data Science and Analytics Program, OU, Dec. 2017.
   Thesis: Dynamic uplift modeling.
3. Pauline M. Ribeyer: Industrial and Systems Engineering, OU, Dec. 2017.
   Thesis: Finding key characteristics of promising compounds for anticancer drug discovery.
4. Samineh Dehghan Nayeri: Industrial and Systems Engineering, OU, Dec. 2017.
   Thesis: Decomposition algorithm for fixed charge time-space network flow problems.
5. Maze M. Enora: Industrial and Systems Engineering, OU, Aug. 2017.
   Thesis: Relationship between the behaviors of social media and physical infrastructure after disruption.
6. Alexandra L. Amidon: Data Science and Analytics Program, OU, May 2017.
   Thesis: A new approach adapting neural network classifiers to sudden changes in nonstationary environments.
7. Emily K. Grimes: Data Science and Analytics Program, OU, May 2017.
   Thesis: The effect of gathering on sandbox player engagement as defined using analytic methods.
8. Snelling E. Megan: Industrial and Systems Engineering, OU, May 2016.
   Thesis: Model for mitigating economic and social disaster damage through structural reinforcement.
9. Abdulrahman Khamaj: Industrial and Systems Engineering, OU, May 2015.
   Thesis: Motorcycle rider performance and distractibility.

M.S. Advising: M.S. Non-Thesis Committee Member (Total: 24)

1. Bhaskar, Muthukrishnan, Industrial and Systems Engineering, OU, Dec 2019.
2. Uchenna Kelvin Egwu, Industrial and Systems Engineering, OU, Dec 2019.
3. Abigail Mace, Industrial and Systems Engineering, OU, 2019.
4. Christopher Carter, Industrial and Systems Engineering, OU, 2019.
5. Ganapathy Appan Kallidaikurichi, Industrial and Systems Engineering, OU, 2017.
6. Karthikeyan Rajendran, Industrial and Systems Engineering, OU, 2017.
7. Nauman Khan, Industrial and Systems Engineering, OU, 2017.
8. Sai Saketh Boyanapalli, Industrial and Systems Engineering, OU, 2017.
9. Abdulshaheed AlBukheder, Industrial and Systems Engineering, OU, 2016.
10. Adnan Alimahomed, Industrial and Systems Engineering, OU, 2016.
11. Akhil Prasanna Degala, Industrial and Systems Engineering, OU, 2016.
12. Bharadwaj Alaha Srimath, Industrial and Systems Engineering, OU, 2016.
13. Gautam Anuradha Muruganandham, Industrial and Systems Engineering, OU, 2016
14. Karnakar Surabhi, Industrial and Systems Engineering, OU, 2016.
15. Matthew Schlegel, Industrial and Systems Engineering, OU, 2016.
16. Naresh Chivukula, Industrial and Systems Engineering, OU, 2016.
17. Nayan Kumar Raguluri, Industrial and Systems Engineering, OU, 2016.
18. Ponsenthil Sivasubramanian, Industrial and Systems Engineering, OU, 2016.
19. Sai Krishna Bhavaraju, Industrial and Systems Engineering, OU, 2016.
20. Sidhaarth Krishnamoorthy, Industrial and Systems Engineering, OU, 2016.
21. Somia Tripathy, Industrial and Systems Engineering, OU, 2016.
22. Srujanveer Reddy Kallem, Industrial and Systems Engineering, OU, 2016
23. Vykram Vijayasekaran, Industrial and Systems Engineering, OU, 2016.
24. Oluwafemi Oseni, Industrial and Systems Engineering, OU, 2014.

Undergraduate Research Assistants (Total: 11)

1. Sierra J. Sallee, Computer Science, OU, Fall 2019. (Honor Eng. Research Experience)
2. Miriam E. Ford, Industrial and Systems Eng., OU, Fall 2019. (Honor Eng. Research Experience)
3. Josh K. Kirby*****, Industrial and Systems Eng., OU, Fall 2019. (Honor Eng. Research Experience)
4. Ashley Jang*****, Economics, Wellesley College, Wellesley, MA, Summer 2019.
5. Jamie C. Jezier, Industrial and Systems Eng., OU, Fall 2018.

6. Ricardo P. Fraga, Industrial and Systems Eng., OU, Spring 2018 and Summer 2018.
7. Lauren N. Yeagle*****, Industrial and Systems Eng., OU, Spring 2017 and Summer 2017.
8. Mattlyn R. Dragoo*****, Industrial and Systems Eng., OU, Spring 2017 and Summer 2017.
9. Morgan E. Corn*****, English Literature, Fall 2016 - Fall 2018.
10. Brooke Brubaker, Industrial and Systems Eng., OU, Summer 2016.
11. Catherine A. Ha*****, Computer Science, Fall 2015 and Spring 2016. (Honor Eng. Research Experience)

****** *Ashely is a student who graduated Norman High School in OK as a top student to join Wellesley.*
****** *Morgan received B.B.A in supply chain management in Fall 2018 and is working at ArcBest in Arkansas.*
****** *Catherine worked as a RA based on my Junior Faculty Fellowship Award and is a co-founder of her own consulting company.*

PROFESSIONAL SERVICE

Membership

2017-current Member of Institute of Industrial and Systems Engineers (IISE).
2012-current Member of IEEE System, Man, and Cybernetics Society (IEEE SMC).
2009-current Member of Human Factors and Ergonomics Society (HFES).

Offices Held

2018-current Editorial Board member of the journal Human Factors and Ergonomics in Manufacturing and Service Industries.
2018-current Editorial Board member of the International Journal of Industrial and Operations Research.
2018-2020 Program Chair of the Education Technical Group in the Human Factors and Ergonomics Society.
2018 Guest Academic Editor of the journal PLOS ONE (International Journal of Information Technology and Decision Making).
2016-2018 Program Chair Elect of the Education Technical Group in the Human Factors and Ergonomics Society.
2014-2016 Student Awards and Affairs Officer of the Cognitive Engineering and Decision Making Technical Group in the Human Factors and Ergonomics Society.

Advisor

2014-current Faculty advisor*****, Human Factors and Ergonomics Society OU Chapter.
*****Dr. Edward Cokely in the Department of Psychology has joined as co-advisor since 2017.

Proposal Reviewer

2020 National Science Foundation (NSF) medium size project ($500K-$1.2M) reviewer: Invited as a panel to review proposals submitted to Cyber Human Systems (CHS) program, Computer and Information Science and Engineering (CISE) division at NSF. (Feb 20-21).

Journal Reviewer

2019-current Reviewer for the Journal of Weather, Climate, and Society within the American Meteorological Society (AMS).
2019-current Reviewer for the IEEE Transactions on Computational Social Systems
2019-current Reviewer for the Journal of Entropy.
2019-current Reviewer for the International Journal of Environmental Research and Public Health
2019-current Reviewer for the Journal of Electronics
2019-current Reviewer for the Journal of Information

2018-current Reviewer for the International Journal of Aerospace Psychology
2018-current Reviewer for the Public Library of Science (PLOS ONE)
2018-current Reviewer for the International Journal of Environmental Research and Public Health
2016-current Reviewer for IEEE Transactions on Affective Computing.
2016-current Reviewer for Applied Ergonomics.
2016-current Reviewer for Ergonomics.
2016-current Reviewer for Human Factors and Ergonomics in Manufacturing and Service Industries.
2014-current Reviewer for Human Factors: The Journal of Human Factors and Ergonomics Society.
2013-current Reviewer for IEEE Transactions on Systems, Man, & Cybernetics.
2013-current Reviewer for IEEE Transactions on Human-Machine Systems.
2013-current Reviewer for IEEE International Conference on Systems, Man, & Cybernetics.
2011-current Reviewer for Human Factors and Ergonomic Society (HFES) Annual Meeting.
2013 Reviewer for International Journal of Human Computer Interaction (HCI).
2011-2012 Reviewer for Journal of Operational Research Society.

Organizing committee member

2019 Session chair. Education technical group lecture session (E3) at the 2019 Human Factors and Ergonomics Society 63rd Annual Meeting, Oct. 28- Nov. 1, Seattle, WA.
2018 Session chair: Usability, assessment, and institutional futures topic session (E4) at the 2018 Human Factors and Ergonomics Society Annual Meeting, Oct 1-5, Philadelphia, PA.
2018 Session chair: Human factors topic session (CHFE-2) at the 2018 Institute of Industrial and Systems Engineers (IISE) Annual Conference, May 19-22, Orlando, FL.

Services at the University of Oklahoma

2019-current Initiated community outreach activity: Collaborating with OU libraries Emerging Technologies office and Thick Descriptions outreach organization to introduce the state-of-the-art virtual reality learning to underrepresented minorities and those in rural, under-developed areas in Oklahoma.
2019-current OU campus security authority: Assisting students on reporting crimes or misconducts.
2019-2020 Initiated joint Human Factors and Ergonomics Society chapter meetings between the University of Oklahoma and the Ohio State University.
2019-2020 Faculty Sponsor for the “Walk to Defeat Amyotrophic Lateral Sclerosis (ALS)” (also known as the Lou Gehrig's disease), student led program.
2018-2020 Faculty Search Committee member for the “ISE Health and Medical Systems.”
2017-current Member of the Research Initiatives in “Psychology, Human Factors, and Forensics” team to develop research topics and functional requirements for National Environment Simulation and Testing (NEST) facility. Supported by the Office of the Vice President for Research.
2017-current Institutional Co-PI of the Federal Aviation Administration Center of Excellence at OU.
2017-2018 ISE Director (i.e. chair) Search Committee member.
2017-current Supporter of Women in Engineering Program.
2016-current Faculty advisor of ISE homepage interface design and content management.
2016-current Invited visiting scholars to OU: Dr. Kwon at the Department of Industrial and Management Engineering at Inje University (in 2016) and Dr. Lee at the Department of Energy Resources Engineering at Pukyong National University (in 2018).
2016-2017 Task force team member for SWOT analysis and the development of strategic research focus areas for the “State and Future of the School of the ISE.”

2014-current ISE Graduate Committee member.
2014-current Developed courses and reports following the based on the Accreditation Board for Engineering and Technology (ABET) criteria.
2014-current Advisor of senior undergraduate Capstone projects (total of 12 projects).

Faculty Development Activities

2018 National Science Foundation (NSF) CAREER Proposal Writing Workshop, NSF Division of Computer, Information Science, and Engineering (CISE) (Apr. 9), NSF Headquarters, Alexandria, VA.
2017 NSF CAREER Proposal Writing Workshop, NSF Division of Civil, Mechanical, Manufacturing and Innovation (CMMI) (Apr. 2-4), Oregon State University, Portland, OR.
2017 Canvas (learning management system) training (Jan. 3-6), Norman, OK.
2016 Center of teaching excellence: Faculty learning community - Creating and implementing course videos (Mar. 23 - Apr. 19), Norman. OK.
2015 Faculty Colloquium, Institute of Industrial Engineers (IIE) conference (May 23-24), Nashville, TN.
2014 New faculty seminar series (Sep. 2 - Nov. 18), Norman, OK.

Media Contributions, Appearances, and Interviews

2020 Brochure (both printed and online) were sent out to 130 Universities to introduce the contents of my NSF CAREER award.
2019 Appeared in Federal Aviation Administration Moroney News (Center Buzz, Vol. 5, Issue 3) for presenting at the Federal Aviation Administration Summit, Mike Monroney Aeronautical Center, OKC.
Article title: Second Annual Human Systems Integration Summit a Success (May, 2019)
Occasion: Presentation of "Improving candidate training through the Universal Design for Learning (UDL) pedagogy."
Archived access: https://www.esc.gov/monroneynews/archive/Vol_5/CB/03_4.asp
2018 Appeared in headlines of the Society of Petroleum Engineers website (www.spe.org)
Article title: Cognitive study of human factors for safe drilling operations: Eye-tracking technology. (Sep., 2018)
Occasion: Interdisciplinary collaboration research using eye tracking technology.
Archived access: https://www.spe.org/en/print-article/?art=4587
2018 Appeared in Federal Aviation Administration Monroney News (Technical Training Focus).
Article title: Technical Training Augments Reality (Aug., 2018)
Occasion: Virtual reality research collaboration with Federal Aviation Administration Civil Aerospace Medical Institute (AAM520) .
Archived access: https://my.faa.gov/focus/articles/2018/07/Technical_Training_A.html
2017 Appeared in IEEE Systems, Man, and Cybernetics Magazine, Issue 2, pp. 52-53.
Extended abstract title: An eye movement analysis algorithm for a multi-element target tracking task: Maximum transition-based agglomerative hierarchical clustering.
Occasion: Andrew P. Sage Best Transactions Paper Award.
Access to article: https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=7902225

GRANTS, COOOPERATIVE AGREEMENTS, CONTRACTS, AND OTHER AWARDS.

Center Awards: Total of $5.9M

Role	Start	End	Minimum Budget	Sponsor	Title
Institutional Co-PI	8/12/16	8/11/21	$5,875,990	Federal Aviation Administration (FAA)	Center of Excellence for Technical Training and Human Performance (COE TTHP): Solutions for Operational Aviation Research (SOAR)

Other External Awards: Total of $1,550,462 (PI share: $1,198,442)

Role	Start	End	Budget	Sponsor	Title
PI	8/15/20	8/14/25	$401,714 (share 100%)	NSF CAREER (CISE: Cyber Human Systems)	CAREER: Non-text-based smart learning in fully immersive multi-person virtual reality using multimodal analysis of physiological measures
PI	9/26/19	9/26/20	$134,153 (share 100%)	FAA Aviation Research Grants Program (AAQ-610) FAA Cooperative Agreement 692M151940008. Project No. A20-0163.	Gamification of visual scanning strategies
PI	4/1/18	1/14/21	$370,324 (share 100%)	FAA Aviation Research Grants Program (AAQ-610) FAA Cooperative Agreement 692M15184001, Project No. A18-0232.	Analysis of human behaviors and strategies for risk reduction: Application in high fidelity virtual reality simulation of real airports
Co-PI (S. Salehi, PI)	10/1/16	9/30/18	$383,271 (share 30%)	National Academies of Science (NAS) 200007356, Project No. A18-0001.	Virtual reality offshore operations training infrastructure: Enhancing expert containment, decision making, and risk communications
PI	9/9/16	12/31/18	$186,000 (share 57%)	FAA Center of Excellence for TTHP FAA Cooperative Agreement 16-C-TTHP-OK-008 Amend #4, Project No. A17-0161.	Universal Design for Learning (UDL) and multi-modal training
PI	9/13/16	2/12/18	$75,000 (share 95%)	FAA Center of Excellence for TTHP FAA Cooperative Agreement 16-C-TTHP-OK-008 Amend #8, Project No. A17-0162.	Characterization and application of air traffic controllers' visual search patterns and control strategies for efficient and effective training
PI	3/11 /15	12/31/16	$49,545 (share 100%)	FAA Aviation Research Grants Program (AAQ-610) FAA Cooperative Agreement 15-G-006, Project No. 15-P-0009.	Development of design concepts and algorithms in eye tracking research for a multi-element objects tracking and controlling task to support human performance analysis

Internal Awards: Total of $10K

Role	Start	End	Budget	Sponsor	Title
PI	8/15/15	5/14/16	$7,000 (share 100%)	*****Office of the Vice President for Research	Development of a mixed methods approach to inform the design of new learning methods for a complex task
PI	8/15/15	5/14/16	$3,000 (share 100%)	Honors Research Assistant Program	Visual scanpath analysis using eye tracking data

*******Received as Junior Faculty Fellowship Award*